

magicmaus tutorial — maltose-binding protein (real data)

The [dummy walkthrough](#) showed the mechanics on 8 methyls. This one runs the same pipeline on a **real, full-size dataset**: the 370-residue *E. coli* maltose-binding protein (MBP), **192 methyls**, using experimental chemical shifts from **BMRB entry 7114** and the crystal structure **PDB 1ANF**. By the end you will have reproduced the headline result — magicmaus committing a single answer for all 192 peaks while never dropping the truth from its envelope — and the three-way comparison against MAGIC and MAUS on identical data.

Prerequisites: `.venv` with `python-sat` and `numpy` (see [README.md](#)). All commands run from the repo root.

1. The dataset

`examples/mbp/` ships the real inputs:

file	what it is
<code>1ANF.pdb</code>	MBP crystal structure
<code>hmqc.tsv</code>	192 methyl HMQC peaks (anonymous ids P1...P192) — the input
<code>hmqc_true.tsv</code>	answer key (BMRB assignment of each peak)
<code>noesy.tsv</code>	simulated 3D (H)CCH NOE network (nearest-8 within 8 Å), boolean
<code>hmqc_tentative.tsv</code>	same HMQC but with 24 peaks pre-anchored (used in §4)

192 methyls across 6 labelled types: Ala 44, Ile 22, Leu 60, Met 6, Thr 20, Val 40. That is a realistic large-protein methyl assignment problem.

2. First run — boolean NOESY

```
.venv/bin/python magicmaus.py \
  examples/mbp/1ANF.pdb examples/mbp/hmqc.tsv examples/mbp/
  noesy.tsv \
  --truth examples/mbp/hmqc_true.tsv --tol-h 0.01 --tol-c 0.05 \
  --out mbp_calls.tsv
```

```
methylys(G nodes)=192  HMQC peaks=192  NOESY cross peaks=1650
NOE match (tol H+-0.01/C+-0.05): firm=502 ambiguous(dropped)=1148
unmatched=0
```

```
--- MAUS envelope (never excludes truth) ---
```

```
unique(1 option)      = 51/192
```

```
ambiguous(2-3 options)= 81/192
```

```
ambiguous(>3 options) = 60/192
```

```
--- magicmaus commitment (single coherent call) ---
```

```
confidence: unique=51 scored=67 ambiguous=74
```

```
--- scored vs truth ---
```

```
truth in MAUS option set = 192/192 = 100.0% (never-exclude
guarantee preserved)
```

```
magicmaus single call    = 121/192 = 63.0% correct
```

```
  unique    : 51/51 = 100.0% correct
```

```
  scored    : 44/67 = 65.7% correct
```

```
  ambiguous: 26/74 = 35.1% correct
```

Read it top to bottom:

- The SAT layer resolves **51/192 peaks uniquely** (all correct) and leaves 141 as multi-option sets — but **the truth is in the option set for all 192** (truth in MAUS option set = 100 %). That envelope guarantee never breaks.
- On top of that, magicmaus **commits a single answer for all 192 peaks: 63.0 % correct**, tiered by confidence (51 unique, 67 scored, 74 ambiguous). That is the number to beat with better data (§3–4).

The per-peak table

```
head -9 mbp_calls.tsv | column -t -s $'\t'
```

label	res_type	call	confidence	n_options	options
truth	call_correct	truth_in_set			
P1	L	L7CD2	scored	2	L7CD1,L7CD2
L7CD1	0	1			
P2	L	L7CD1	scored	2	L7CD1,L7CD2
L7CD2	0	1			
P3	V	V8CG1	ambiguous	2	V8CG1,V8CG2
V8CG1	1	1			
P4	V	V8CG2	scored	2	V8CG1,V8CG2
V8CG2	1	1			
P5	I	I9CD1	unique	1	I9CD1
I9CD1	1	1			

P6	I	I11CD1	unique	1	I11CD1
I11CD1	1	1			
P7	L	L20CD1	scored	2	L20CD1, L20CD2
L20CD1	1	1			
P8	L	L20CD2	ambiguous	2	L20CD1, L20CD2
L20CD2	1	1			

P5/P6 are pinned uniquely (unique, certain). P1/P2 are a Leu geminal pair the boolean network cannot orient, so magicmaus swaps them here — but note `truth_in_set = 1` for both: the right answer is still inside each envelope. This is exactly the boolean coin-flip the intensity column fixes next.

3. Add intensities

Real NOE cross peaks carry an intensity $\sim 1/r^6$ — the “how close” signal a boolean edge lacks. `make_intensity_noesy.py` reconstructs it for this dataset (from the truth key + structure), simulating a real experiment:

```
.venv/bin/python make_intensity_noesy.py examples/mbp/1ANF.pdb \
    examples/mbp/hmqc_true.tsv examples/mbp/noesy.tsv
    mbp_noesy_intensity.tsv

.venv/bin/python magicmaus.py examples/mbp/1ANF.pdb examples/mbp/
    hmqc.tsv \
    mbp_noesy_intensity.tsv --truth examples/mbp/hmqc_true.tsv \
    --tol-h 0.01 --tol-c 0.05 --soft-ambiguous
```

NOESY	single-answer correct	scored-tier correct
boolean	63.0 %	65.7 %
+ intensities	87.0 %	92.3 %
+ intensities --soft-ambiguous	87.5 %	92.6 %

`--soft-ambiguous` folds in the ambiguous NOE cross peaks MAUS discards as extra, intensity-weighted tie-breakers. Intensities lift the run **63 % → 87 %** — and the envelope stays at **100 %** the whole way. The scoring layer, idle on boolean data, does real work once the data carry information: with the intensity column in hand the NOE objective’s optimum is the truth, and magicmaus’s 3-cycle annealer climbs to it (a plain greedy ascent stalls ~15 % short).

4. Two more levers

Tighter matching. Sharper shift tolerances make more NOEs resolve firmly, so more peaks land in the unique tier:

```
.venv/bin/python magicmaus.py examples/mbp/1ANF.pdb examples/mbp/
    hmqc.tsv \
    examples/mbp/noesy.tsv --truth examples/mbp/hmqc_true.tsv \
    --tol-h 0.01 --tol-c 0.02
# unique(1 option) = 70/192 ; single call = 156/192 = 81.2%
    correct
```

Tentative anchors. If you already know a few assignments, write them into the res_type cell of the HMQC file (e.g. L45D2 instead of L). hmqc_tentative.tsv carries 24 such anchors; they propagate through the NOE network and sharpen the rest:

```
.venv/bin/python magicmaus.py examples/mbp/1ANF.pdb \
    examples/mbp/hmqc_tentative.tsv examples/mbp/noesy.tsv \
    --truth examples/mbp/hmqc_true.tsv --tol-h 0.01 --tol-c 0.05
# unique(1 option) = 79/192 ; single call = 136/192 = 70.8%
    correct
```

24 anchors lift the unique tier from 51 to 79 — a few knowns go a long way.

5. The payoff — MAGIC vs MAUS vs magicmaus on identical data

convert_to_magic.py re-emits the intensity dataset as a MAGIC control bundle, so all three engines are scored on the **same** shifts, structure and NOE peaks. score_three.py runs the comparison:

```
.venv/bin/python convert_to_magic.py examples/mbp/hmqc.tsv \
    mbp_noesy_intensity.tsv examples/mbp/1ANF.pdb
    magic_run_intensity
# then run MAGIC on magic_run_intensity/control.txt (needs a ../
    magic checkout),
# and:
.venv/bin/python score_three.py
```

	methy-level	residue-level	
truth-in-envelope			
MAGIC (scoring)	11/192 = 5.7%	20/192 = 10.4%	—
(no envelope)			
MAUS (SAT)	unique 51/192 = 26.6%		
192/192 = 100.0%			
magicmaus	167/192 = 87.0%	173/192 = 90.1%	
192/192 = 100.0%			

magicmaus +soft-amb 168/192 = 87.5% 172/192 = 89.6%
192/192 = 100.0%

Two facts the shared network makes plain:

- **MAUS cannot use intensity** — its constraints are boolean, so the 5th column changes nothing (identical 51 unique / 100 % envelope). It bounds the truth but stays undecided on the degeneracy.
- **MAGIC uses intensity but still lands at ~6–10 %** — global scoring over the full candidate space sits on a near-flat landscape and commits to a residue-level answer that cannot even resolve geminal pairs.

magicmaus runs the *same* intensity-weighted scoring MAGIC uses, but only inside MAUS's truth-containing domains — landing **15× MAGIC's methyl accuracy while keeping MAUS's 100 % envelope**. That is the synthesis: MAUS keeps the truth in reach, MAGIC's scoring extracts every bit of experimental signal to commit correctly within it. Neither half delivers this alone.

6. What to keep from a real run

On data without a truth key, drop `--truth` and read `mbp_calls.tsv`:

- **unique** calls — forced by hard constraints; trust them.
- **scored** calls — the NOE geometry's best pick; strong leads, verify where it matters.
- **ambiguous** calls — carry the *whole option set*, not the single call; these are genuine symmetries (mostly geminal pairs) awaiting stereo labelling, a 4D experiment, or more anchors.

The one invariant across every setting above: **the true methyl is never dropped from a peak's option set**. magicmaus commits when it can and abstains — visibly — when it cannot.

7. Beyond MBP — the five-protein benchmark

MBP is one of five targets. `make_peaklists.py` builds the same peak lists from any PDB + BMRB deposition, so the four de-novo blind targets of the MAUS paper (IL-2 and the Cas9 HNH/REC2/REC3 domains) drop straight in — see [../README.md#benchmark--five-real-bmr-b-targets](#) and each `examples/<target>/README.md`. Across all five the **100 % never-exclude envelope holds**, and magicmaus beats full-space MAGIC by up to ~15× while committing on every peak.